Day 2: Relativistic Kinematics I & II: Rapidity & Pseudorapidity in Detail

14-Day Hands-on Course on Heavy-Ion Phenomenology

Lecture

NIT-J, HEP-119

June 2026

Syllabus: 14-Day Hands-on Course Layout

Part I: Relativistic Kinematics

- Day 1: Foundations of Relativistic Kinematics & Experimental HEP
- **Day 2: Kinematics I & II: Rapidity & Pseudorapidity in Detail**
- Day 3: Kinematics III: Invariant Yield & Phase Space Jacobian
- Day 4: Kinematics IV: Coordinate Calculator implementations

Part II: AMPT Physics & Mechanics

- Day 5: AMPT Evolution: HIJING, ZPC, Hadronization, ART
- Day 6: AMPT Data format: event headers & parsing
- Day 7: Single-Particle Observables: Multiplicity & p_T

Part III: Thermodynamics & Collectivity

- Day 8: Transverse Spectra, Identified Particles & m_T Scaling
- Day 9: Centrality & Geometry: Glauber vs AMPT
- Day 10: Collective Flow: Elliptic Flow (v_2)
- Day 11: Medium Modification: ZPC Knobs & Mean Free Path

Part IV: Advanced Scans & Pipeline

- Day 12: Energy Scan: Beam Energy Scan (STAR at RHIC)
- Day 13: Advanced Analysis
- Day 14: Wrap-Up: Reproducible HEP Data Pipeline

Polar Angle to Pseudorapidity Mapping

Theoretical Definition (Sahoo Eq. 5.138)

At ultra-relativistic limits ($p \gg m_0$), the momentum rapidity y simplifies to a purely angular, geometric quantity called **Pseudorapidity** (η):

$$\eta = -\ln \tan\left(\frac{\theta}{2}\right)$$

Where θ is the polar angle relative to the beam axis (z).

Physical Advantages of Pseudorapidity

- **No Particle ID (PID) needed:** Calculated solely from the emission angle θ of the trajectory in the tracker.
- **No magnetic field needed:** Does not require momentum information (only spatial tracking lines).
- **Symmetric / Odd behavior:** It is odd about $\theta = 90^\circ$ ($\eta = 0$). That is, $\eta(180^\circ - \theta) = -\eta(\theta)$, reflecting forward-backward symmetry.

Angular Detector Mapping: θ vs. η

Angle to η Reference (Sahoo Table 5.3)

Polar Angle θ	Pseudorapidity η	Detector Region
0°	∞	Forward Beam pipe
10°	2.44	Endcap region
30°	1.32	Forward transition
45°	0.88	Diagonal transition
90°	0.00	Central Barrel
135°	-0.88	Diagonal transition
170°	-2.44	Backward transition
180°	$-\infty$	Backward Beam pipe

Physical Trajectories (Sahoo Fig. 5.13)

- $\theta = 90^\circ$ ($\eta = 0$): Particles fly straight out transversely.
- $\theta \to 0^\circ$ ($\eta \to \infty$): Particles fly forward along the beam pipe.
- High $|\eta|$ values denote regions extremely close to the beam line (demanding high spatial resolution trackers).

Interactive Physics Simulator

Open in your browser:

- `animations/05_pseudo-rapidity_detector_angle.html`

(Interactive θ vs. η detector mapping)

Coordinate Transformation: From (y, p_T) to (η, p_T)

Vector Decompositions (Sahoo Eq. 5.141 – 5.143)

By defining the pseudorapidity η in terms of longitudinal momentum p_z and total momentum magnitude $|\mathbf{p}|$:

$$e^{\eta} = \sqrt{\frac{|\mathbf{p}| + p_z}{|\mathbf{p}| - p_z}}, \quad e^{-\eta} = \sqrt{\frac{|\mathbf{p}| - p_z}{|\mathbf{p}| + p_z}}$$

We can solve for the momentum component variables in terms of (p_T, η):

$$|\mathbf{p}| = p_T \cosh\eta, \quad p_z = p_T \sinh\eta$$

Which allows us to write the energy E and transverse mass m_T elegantly:

$$E = \sqrt{|\mathbf{p}|^2 + m_0^2} = \sqrt{p_T^2 \cosh^2\eta + m_0^2}$$

The Phase-Space Jacobian: Mathematical Definition

Rapidity to Pseudorapidity Mapping (Sahoo Eq. 5.147)

The distribution of particles as a function of pseudorapidity ($dN/d\eta$) is related to the rapidity distribution (dN/dy) via the phase-space Jacobian $J(y \to \eta) = \mathrm{d}y/\mathrm{d}\eta$:

$$\frac{\mathrm{d}N}{\mathrm{d}\eta \mathrm{d}p_T} = J(y \to \eta) \frac{\mathrm{d}N}{\mathrm{d}y \mathrm{d}p_T} = \sqrt{1 - \frac{m_0^2}{m_T^2 \cosh^2 y}} \frac{\mathrm{d}N}{\mathrm{d}y \mathrm{d}p_T}$$

In the ultra-relativistic (massless) limit $m_0 \to 0$, the Jacobian simplifies to the particle's longitudinal velocity: $J(y \to \eta) \to p/E = \beta \to 1$.

Integration Over Transverse Momentum

To obtain the overall multiplicity distribution $\mathrm{d}N/\mathrm{d}\eta$, we must integrate over the transverse momentum p_T:

$$\frac{\mathrm{d}N}{\mathrm{d}\eta} = \int_0^\infty J(y \to \eta, p_T) \frac{\mathrm{d}N}{\mathrm{d}y \mathrm{d}p_T} \mathrm{d}p_T$$

Because J depends on p_T (through $m_T = \sqrt{p_T^2 + m_0^2}$), the mapping varies strongly with the transverse momentum of the tracks.

The Phase-Space Jacobian: Physical Consequences

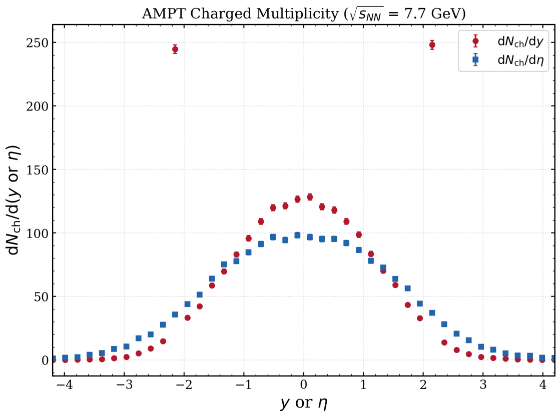
Physical Limits of the Jacobian Factor

- **Massless limit ($m_0 \to 0$ or $p \gg m_0$):** $J(y \to \eta) \to 1 \implies \mathrm{d}N/\mathrm{d}\eta \approx \mathrm{d}N/\mathrm{d}y$. This is highly accurate for photons, electrons, or very high p_T hadrons (where mass is negligible).
- **Massive limit ($m_0 > 0$):** $J(y \to \eta) < 1$ at mid-rapidity ($y \approx 0$). This creates a local, kinematic suppression in the pseudorapidity distribution $\mathrm{d}N/\mathrm{d}\eta$ even when the underlying physical rapidity distribution $\mathrm{d}N/\mathrm{d}y$ is flat or peak-like.

Takeaway

The difference between dN/dy and $dN/d\eta$ is a purely mathematical consequence of the change of variables. Detector trackers measure η directly, but physics analyses must correct back to y using the Jacobian factor to access the true dynamics.

AMPT: $\mathrm{d}N/\mathrm{d}y$ vs. $\mathrm{d}N/\mathrm{d}\eta$ at $\sqrt{s_{NN}} = 7.7$ GeV



Symmetric Collision at $\sqrt{s_{NN}} = 7.7$ GeV

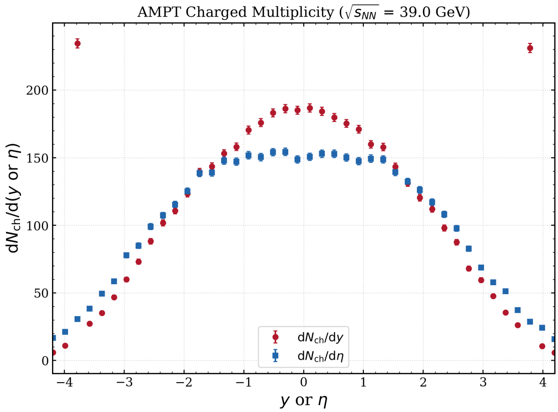
- Represents the lower energy regime of the RHIC Beam Energy Scan.
- The rapidity range is restricted ($y_{\mathrm{beam}} \approx 2.09$), resulting in a narrow distribution.
- Peak charged particle yield per event is relatively small ($\mathrm{d}N/\mathrm{d}y \approx 45$).
- **Jacobian Suppression:** The suppression at $\eta = 0$ is visible as a flattening of the pseudorapidity peak compared to the rapidity distribution.

Interactive Jacobian Explorer

Open in your browser:

- `day2/day2_rapidity_jacobian_explorer.html`

AMPT: $\mathrm{d}N/\mathrm{d}y$ vs. $\mathrm{d}N/\mathrm{d}\eta$ at $\sqrt{s_{NN}} = 39$ GeV



High Energy Collision at $\sqrt{s_{NN}} = 39$ GeV

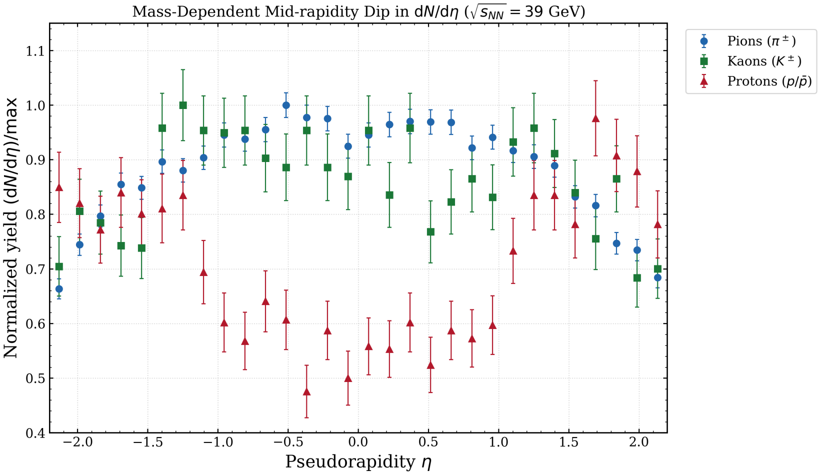
- High energy regime with a wider longitudinal phase space ($y_{\mathrm{beam}} \approx 3.73$).
- Peak yield is significantly higher ($\mathrm{d}N/\mathrm{d}y \approx 120$) due to increased particle production.
- The difference between dN/dy and $dN/d\eta$ is more pronounced across a broader range, showing a wide plateau in pseudorapidity.

Interactive Jacobian Explorer

Open in your browser:

- `day2/day2_rapidity_jacobian_explorer.html`

Mass-Dependent Mid-rapidity Dip in $\mathrm{d}N/\mathrm{d}\eta$



Mass-Dependent Mid-rapidity Dip: The Jacobian Mechanism

Why do heavy particles dip at $\eta = 0$?

At mid-rapidity ($\eta = 0 \iff y = 0$), the Jacobian simplifies to: $J(0) = \frac{p_T}{m_T} = \frac{p_T}{\sqrt{p_T^2 + m_0^2}}$.

- **Pions ($\pi^{\pm}$, $m_0 \approx 0.14$ GeV):** Very light mass. The ratio $p_T/m_T \approx 1$ for almost all p_T. Thus, $J(0) \approx 1$ (no dip).
- **Kaons ($K^{\pm}$, $m_0 \approx 0.49$ GeV):** Medium mass. Suppression at low p_T is significant, creating a shallow dip.
- **Protons ($p/\bar{p}$, $m_0 \approx 0.94$ GeV):** Heavy mass. Suppression is severe ($J(0) \ll 1$ for $p_T < m_0$), creating a deep, pronounced mid-rapidity dip.

Physical Takeaway

This dip is purely a coordinate mapping artifact, not physical dynamics (since dN/dy remains flat or peak-like at $y = 0$). Heavy particles have a large rest mass, making their velocity $\beta \ll 1$ at low p_T, which suppresses $dN/d\eta$ via $J \to \beta$.

Rapidity Boosts: Lab Frame vs. Center-of-Mass Frame

Rapidity Additivity (Sahoo Eq. 5.43)

Under a collinear Lorentz boost of velocity β_b (boost rapidity y_b), the rapidity transforms linearly: $y' = y - y_b$. This shifts the entire distribution $\mathrm{d}N/\mathrm{d}y'$ as a rigid body, preserving its shape and width σ_y.

Relativistic Beaming (Searchlight Effect)

At high boosts, particles are beamed forward (in the direction of the boost). This increases the local density at high η', showing polar angle bunching.

Pseudorapidity Deformation

Pseudorapidity η does **not** transform additively under boosts because the polar angle θ transforms non-linearly. The distribution $\mathrm{d}N/\mathrm{d}\eta'$ skews and deforms.

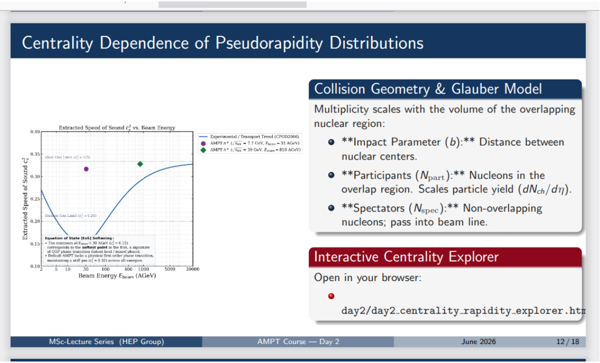
Interactive Boost Simulator

Open in your browser:

- `day2/day2_lab_vs_cms_rapidity.html`

  (Observe rapidity shift vs pseudorapidity deformation and beaming)

Centrality Dependence of Pseudorapidity Distributions



Collision Geometry & Glauber Model

Multiplicity **amplitude** scales with the overlapping nuclear volume:

- **Impact Parameter (b):** Distance between nuclear centers. Central ($b \approx 0$ fm, $0-5\%$) vs. Peripheral ($b > 12$ fm, $> 70\%$).
- **Participants (N_{part}):** Nucleons undergoing ≥ 1 inelastic collision. Directly sets the fireball volume and multiplicity yield.
- **Spectators (N_{spec}):** Non-overlapping cold nuclear remnants. Continue at beam rapidity into Zero Degree Calorimeters (ZDCs).

Interactive Centrality Explorer

Open in your browser:

- `day2/day2_centrality_rapidity-_explorer.html`

Physics of Centrality & Multiplicity Scaling

Two-Component Multiplicity Scaling

The total charged particle multiplicity at mid-rapidity follows the two-component Glauber model:

$$\frac{\mathrm{d}N_{\mathrm{ch}}}{\mathrm{d}\eta} \approx n_{pp} \left[(1-x)\frac{N_{\mathrm{part}}}{2} + xN_{\mathrm{coll}} \right]$$

Where $x \approx 0.15$ at LHC energies.

- **Soft Processes ($\propto N_{\mathrm{part}}$):** String fragmentation and low-Q^2 QCD processes.
- **Hard Processes ($\propto N_{\mathrm{coll}}$):** Point-like perturbative QCD scatterings (jets, mini-jets), which dominate at high energies.

What Centrality Changes (and What It Does Not)

- **Amplitude $\uparrow$ (central):** More participating nucleons $\Rightarrow$ higher $\mathrm{d}N_{\mathrm{ch}}/\mathrm{d}\eta$. The curves stack vertically.
- **Shape (plateau width) unchanged:** The width σ_y of the rapidity distribution is set by the *beam energy* $\sqrt{s_{NN}}$, not by centrality. Normalized per participant pair, the distributions are self-similar.
- **Jacobian dip shallower in central:** Larger radial flow in central collisions boosts $\langle p_T \rangle$, increasing β and making the Jacobian suppression at $\eta = 0$ slightly less pronounced.

Landau Hydrodynamics & Rapidity Widths

Landau Model Rapidity Distribution (Sahoo pg. 145–146)

In Landau's hydrodynamic model of relativistic expansion, the rapidity distribution of produced pions is described by a Gaussian:

$$\frac{\mathrm{d}N}{\mathrm{d}y} = \frac{N}{\sqrt{2\pi\sigma_y^2}} \exp\left(-\frac{y^2}{2\sigma_y^2}\right)$$

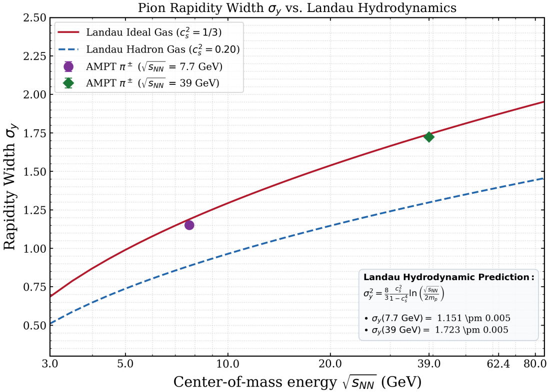
Where the width of the rapidity distribution σ_y relates to the center-of-mass energy and the speed of sound c_s in the medium:

$$\sigma_y^2 = \frac{8}{3}\frac{c_s^2}{1-c_s^4} \ln\left(\frac{\sqrt{s_{NN}}}{2m_p}\right)$$

Landau Model Scaling Limits

- **Ideal Gas limit ($c_s^2 = 1/3$):** Prefactor becomes 1, so $\sigma_y = \sqrt{\ln(\sqrt{s_{NN}}/2m_p)}$.
- **Hadron Gas limit ($c_s^2 \approx 0.20$):** Premature freezing or lower speed of sound shrinks the rapidity width σ_y.

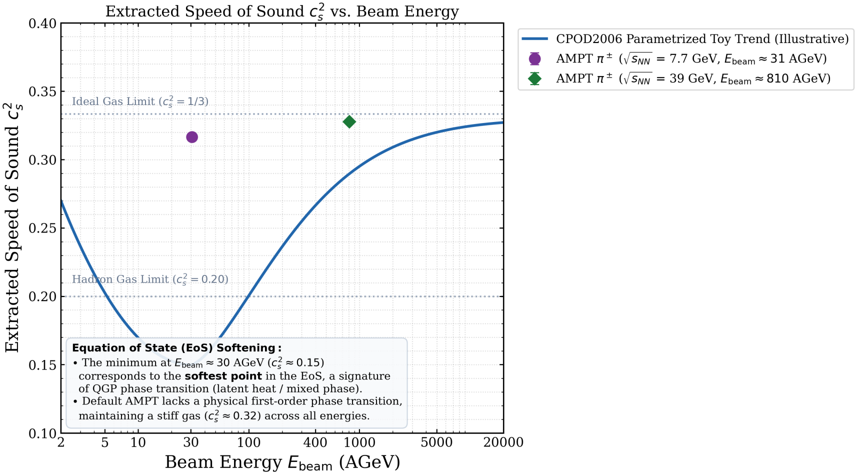
Pion Rapidity Widths: Landau Hydrodynamics



Width σ_y Analysis in AMPT

- The standard deviation σ_y of the pion rapidity distribution is calculated from AMPT at 7.7 and 39 GeV.
- Theoretical curves show Landau predictions for the Ideal Gas ($c_s^2 = 1/3$) and Hadron Gas ($c_s^2 = 0.20$).
- **Observations:** AMPT pion rapidity widths are slightly wider and lie close to the Ideal Gas limit, indicating a stiff equation of state (EoS) in the default AMPT transport model.

Equation of State: Extracted Speed of Sound



Equation of State: Speed of Sound & EoS Softening

Speed of Sound $c_s^2 = \partial P/\partial \epsilon$ (Sahoo Fig. 5.16)

In Relativistic Hydrodynamics, c_s^2 characterizes the stiffness of the early medium. We extract c_s^2 from the pion rapidity width σ_y by inverting the Landau equation:

$$c_s^2 = \frac{-4\ln(\sqrt{s_{NN}}/2m_p)}{3\sigma_y^2} + \sqrt{\left(\frac{4\ln(\sqrt{s_{NN}}/2m_p)}{3\sigma_y^2}\right)^2 + 1}$$

Physical Interpretation & Motivation

- **The "Softest Point":** Experimental data shows a clear minimum ($c_s^2 \approx 0.15$) at $E_{\mathrm{beam}} \approx 30$ AGeV ($\sqrt{s_{NN}} \approx 7.7$ GeV).
- **Transition Signature:** This minimum is a classic signature of the transition from hadronic matter to QGP (mixed phase or onset of deconfinement).
- **AMPT Default:** Remains stiff ($c_s^2 \approx 0.32$), showing the lack of a physical first-order phase transition in default ZPC/ART.

Summary: Day 2 Hands-On Laboratory Exercises

Objective: Analytical & Data-Driven Rapidity Calculations

All exercises are framed in your Jupyter Notebooks located in the `exercises/` folder:

- `exercises/day02_hands_on.ipynb` (Student skeleton notebook)
- `exercises/day02_hands_on_solutions.ipynb` (Reference solutions with code and plots)

1. **Problem 1:** Calculate center-of-mass energy $\sqrt{s_{NN}}$ for RHIC Energy Scan values.
2. **Problem 2:** Prove why collider configurations have a massive kinematic advantage over fixed-target configurations.
3. **Problem 3:** Compute and verify Lorentz factors γ, velocities β, and beam rapidity limits y_{beam} for LHC Pb+Pb beams.
4. **Problem 4:** Parse AMPT events, extract particle species (pions, kaons, protons), and plot the mass-dependent mid-rapidity dip in $\mathrm{d}N/\mathrm{d}\eta$.
5. **Problem 5:** Extract the standard deviation (width σ_y) of pion rapidity and check compatibility with Landau's ideal gas limit vs. viscous gas models.